

# SAFWAN SHIBLEE

Waterloo, ON

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## Education

### University of Waterloo

Expected Graduation 2029

*B.A.Sc. Mechatronics Engineering — GPA 3.7/4.0, President's Scholarship of Distinction*

*Waterloo, ON*

## Experience

### Wolf Advanced Technology

Jan 2026 – Apr 2026

*Software Engineer*

*Aurora, ON*

- Built a **Notifications-as-a-Service (NaaS)** platform in **ASP.NET Core (.NET 10)** with async Email/Teams routing, **fault-tolerant delivery** (3 retries, exponential backoff), and full unit + integration test coverage
- Integrated **Microsoft Graph API** with internal **observability/logging** systems for real-time monitoring across distributed services; enforced structured logging patterns adopted across the team
- Automated PCB engineering toil with **PowerShell/TCL/SKILL** pipelines for DRC validation, footprint manipulation, and batch IPC2581/XML export — eliminating **10+ hours of manual effort per board** across 15+ engineers
- Containerized a web-based PCB viewer (**Docker**, Blazor, JS) parsing IPC2581/XML files; cut load time from ~10 min to 5–7 min, backed by **1,100+ unit tests** over parsing logic and data models

### GLOCAL Foundation of Canada

May 2025 – Aug 2025

*Data Analyst*

*Waterloo, ON*

- Deployed and operated a full-stack election platform (**Docker + React/Leaflet + FastAPI + Supabase/PostgreSQL**) serving **2.4k+ records**; owned backend hosting, schema design, and REST API integration end-to-end
- Built data ingestion + cleaning pipelines in **Python (pandas, NumPy)**; led a 30-participant workshop and coordinated a 25-intern cross-functional team

### Fintech Startup (Under NDA)

Jan 2025 – Dec 2025

*Frontend Developer*

*Waterloo, ON*

- Applied **TDD** with unit + integration tests reaching **70% coverage** and cutting regression bugs by **40%**; integrated **Firestore Auth** and **GCP** cloud backend services into the Flutter client
- Hardened the auth flow and 20+ shared Flutter components, cutting login failures by **30%** and stabilizing iOS/Android release consistency across cross-platform builds

### InI Labs Inc.

Jul 2023 – Aug 2023

*Full Stack Developer*

*Waterloo, ON*

- Maintained a scientific calculator inside a **150,000+ user** production app; resolved 30+ critical bugs and automated backend regression testing, improving cross-device reliability

## Projects

### Self-Healing SRE Monitor + ML Anomaly Detection | *Python, Docker, Terraform, AWS, Prometheus* April 2026

- Architected a **5-container SRE stack** on **AWS EC2** via **Terraform** (Amazon Linux 2, IAM, security groups); auto-detects and recovers from container failures at **<30s MTTR** with zero human intervention
- Engineered a Python health agent polling **/health** every 5s; on 3 consecutive failures opens an **SQLite-backed incident**, restarts the crashed container via the **Docker SDK**, and posts structured failure/recovery alerts to **Slack**
- Built a parallel **ML anomaly detector** running **three Isolation Forest models** (per-metric: RPS, p95 latency, error rate) trained on a Prometheus collected baseline — surfacing degradation *before* SLO breach with  $\sigma$ -attributed alerts
- Shipped a push-to-deploy pipeline via **GitHub Actions** (scp/ssh) rebuilding the EC2 stack in ~2 min on every merge to main; **nginx**-served ops dashboard surfaces live RPS, p95 latency,  $\pm\sigma$  anomaly band, and incidents

### Property Tax Calculator | *React, Next.js, TypeScript, Python, Flask, SQLite*

March 2025

- Built a full-stack web app with a **Flask REST API + SQLite** backend and **React/Next.js** frontend; cut load times by **25%** and redundant API calls by **30%** via memoization and request batching

### Prayer Times Application | *Dart, Flutter, Android (Orange Pi SBC), GitHub Pages*

Oct 2022 – Oct 2024

- Built a cross-platform **Flutter** app (Android APK + Web) deployed on an Orange Pi as the center's **24/7 prayer-times display** for **2 years**, viewed by **700+** congregants
- Engineered for **unattended 24/7 operation** with a self-contained schedule (no API/DB dependency), surviving network outages with **zero-touch ops** across the full 2-year deployment

## Technical Skills

**DevOps & Cloud:** Docker, Terraform (IaC), AWS (EC2/IAM), GCP, Azure DevOps, GitHub Actions (CI/CD), nginx, Linux, Bash, Jira

**Observability & SRE:** Prometheus, Grafana, Isolation Forest anomaly detection, structured logging, Slack alerting, self-healing systems, retry/backoff patterns

**Languages:** Python, JavaScript/TypeScript, C/C++, C#, Java, Dart/Flutter, PowerShell, SQL, TCL, SKILL

**Frameworks & DBs:** ASP.NET Core, Blazor, FastAPI, React/Next.js, Flask, Node.js, PostgreSQL, MySQL, Supabase, Firebase, SQLite